installed inside a guest to improve its performance and extend the functionality. For instance, these allow us to share folders between the host and guest, and to drag and drop functionality.

Features of VirtualBox

Let us discuss some important features of VirtualBox.

1) *Portability:* VirtualBox is highly portable. It is available on a wide range of platforms and its functionality remains identical on each of those platforms. It uses the same file and image format for VMs on all platforms. Because of this, a VM created on one platform can be easily migrated to another. In addition, VirtualBox supports the Open Virtualisation Format (OVF), which enables VM import and export functionality.
2) *Commodity hardware:* VirtualBox can be used on a CPU that doesn't support hardware virtualisation instructions, like Intel's VT-x or AMD-V.
3) *Guest additions:* As stated earlier, these software bundles are installed inside a guest, and enable advanced features like shared folders, seamless windows and 3D virtualisation.
4) *Snapshot:* VirtualBox allows the user to take consistent snapshots of the guest. It records the current state of the guest and stores it on disk. It allows the user to go back in time and revert the machine to an older configuration.
5) *VM groups:* VirtualBox allows the creation of a group of VMs and represents them as a single entity. We can perform various operations on that group like *Start, Stop, Pause, Reset,* and so on.

Getting started with VirtualBox

System requirements

VirtualBox runs as an application on the host machine and for it to work properly, the host must meet the following hardware and software requirements:

1) An Intel or AMD CPU
2) A 64-bit processor with hardware virtualisation is required to run 64-bit guests
3) 1GB of physical memory
4) Windows, OS X, Linux or Solaris host OS

Downloading and installation

To download VirtualBox, visit *https://www.virtualbox.org/wiki/Downloads link.* It provides software packages for Windows, OS X, Linux and Solaris hosts. In this column I'll be demonstrating VirtualBox on Mint Linux. Refer to the official documentation if you wish to install it on other platforms.

For Debian based Linux, it provides the '.deb' package. Its format is *virtualbox-xx_xx-yy-zz.deb* where *xx_xx-yy* is the version and build number respectively and *zz* is the host OS's name and platform. For instance, in case of a Debian based 64-bit host, the package name is *virtualbox-5.2_5.2.0-118431-Ubuntu-xenial_amd64.deb.*

To begin installation, execute the command given below in a terminal and follow the on-screen instructions:

```
$ sudo dpkg -i virtualbox-5.2_5.2.0-118431-Ubuntu-xenial_amd64.deb
```

Using VirtualBox

After successfully installing VirtualBox, let us get our hands dirty by first starting VirtualBox from the desktop environment. It will launch the VirtualBox manager window as shown in Figure 1.

Figure 1: VirtualBox manager

This is the main window from which you can manage your VMs. It allows you to perform various actions on VMs like Create, Import, Start, Stop, Reset and so on. At this moment, we haven't created any VMs; hence, the left pane is empty. Otherwise, a list of VMs are displayed there.

Creating a new VM

Let us create a new VM from scratch. Follow the instructions given below to create a virtual environment for OS installation.

1) Click the 'New' button on the toolbar.
2) Enter the guest's name, its type and version and click the 'Next' button to continue.
3) Select the amount of memory to be allocated to the guest and click the 'Next' button.
4) From this window we can provide storage to the VM. It allows us to create a new virtual hard disk or use the existing one.
   4a) To create a new virtual hard disk, select the 'Create a virtual hard disk now' option and click the 'Create' button.
   4b) Select the VDI disk format and click on 'Continue'.
   4c) On this page, we can choose between a storage policy that is either dynamically allocated or a fixed size:
   i) As the name suggests, a dynamically allocated disk will grow on demand up to the maximum provided size.
   ii) A fixed size allocation will reserve the required storage upfront. If you are concerned about performance, then go with a fixed size allocation.
   4d) Click the 'Next' button.
5) Provide the virtual hard disk's name, location and size before clicking on the 'Create' button.
   This will show a newly created VM on the left pane as seen in Figure 2.